

10th International Conference of Information and Communication Technology (ICICT-2020)

Triple module redundancy reliability framework design based on heterogeneous multi-core processor

Haomiao Su, Tiejun Lu*, Changlei Feng, Lei Chen

Beijing Microelectronics Technology Institute, 2 North Siyingmen Road Donggaodi Street, Beijing, China

Abstract

To solve the problem of commercial device with weak anti-SEU ability and traditional radiation hardened device with slow processing speed, this article proposes a novel system-level collaborative triple module redundancy (TMR) design to mitigate single event upset (SEU) on the advanced heterogeneous processor. Based on Xilinx Zynq Ultrascale+ MPSoC, we combine TMR with rolling back, watchdog and scrubbing technique between heterogeneous cores which are Arm Cortex-A53 and Field Programmable Gate Array (FPGA). This combination brings us a promising result of fault identification and correction according to the simulation of the single channel data fault and the double channel data fault. It costs 27%-44% more time to process error and the system has relatively low resource utilization rate and power consumption. Moreover, it can also apply to general heterogeneous quad-core System on Chip (SoC) with FPGA processor's anti-SEU design and provides some ideas for further study of system level radiation-resistant chips' design.

© 2021 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Peer-review under responsibility of the scientific committee of the 10th International Conference of Information and Communication Technology.

Keywords: TMR, SEU Mitigation, Heterogeneous Processor, Fault Tolerance

1. Introduction

With the booming development of commercial spaceflight and space technology, more and more spacecrafts will be launched into space. High reliable control and high-performance data processing is the key matter of the satellite chip¹. However, all electrical devices within the satellite are in the high radiation environment and the single event

* Corresponding author. Tel.: +86-139-1115-8923.

E-mail address: 13911158923@139.com

upset (SEU) will cause a lot of trouble to influence chips' operation. So anti-SEU is becoming another key matter of space mission.

In recent years, double module redundancy (DMR)² and TMR³ are both used to mitigate SEU. As for DMR, Zynq 7000 serials are used to conduct a DMR system with TMR reinforced register⁴ and DMR voter is also thoroughly studied⁵. As for TMR, ARM uses triple core lock-step, which is a kind of TMR, to reinforce the Cortex-R serial chip and it can apply to communication satellite and autopilot^{6, 7}. Others also use this technique and Cortex-R chip to design a high reliable aircraft control system⁸. About SPARC V8 processor, TMR is also used to mitigate SEU and guarantee it will work correctly in satellite⁹. Moreover, heterogeneous processor with three different framework proves that it has the ability to mitigate error¹⁰ and roll back technique combined with TMR¹¹ is also being used. Besides engineer study, theoretical study of TMR is also confirmed that TMR is suitable for error identification and correction^{12, 13} and relative system level TMR design flow is also being developed¹⁴.

In triode level, triode's threshold voltage drops gradually, which leads to radiation resistant ability decline because of scaling down¹⁵⁻¹⁷. Thus, triode is more vulnerable to SEU and the number of triodes within the effecting area increases. Traditional rad-hard technology^{18, 19} uses peripheral circuit to ensure the triode's function. With the affecting triode's number growing in the radiation area, the peripheral circuit can also be affected, which will weaken the radiation resistance. Moreover, the circuit verification becomes more difficult and the traditional rad-hard design slow down the processing speed due to the redundant circuit²⁰.

Therefore, system level radiation reinforcement is a better option for anti-SEU. In system level reinforcement, chips don't need to be redesigned and DMR or TMR can be applied directly which chip still possesses high process speed. However, DMR can only identify whether the two data are the same. It cannot correct the error. TMR can do both and is more maneuverable which the structure is shown in Figure 1.

In previous SEU mitigation study, TMR design is mainly applied to homogeneous processors because former commercial heterogeneous processor, which is lack of cores or buses, is unsuitable for TMR. With the development, heterogeneous processor's structure is fulfilled the TMR design and the processor has high process speed. However, the previous TMR design is not suitable for heterogenous cores. Thus, we propose a novel collaborative TMR technique to reinforce the advanced commercial heterogeneous multi-core processors' SEU mitigation ability.

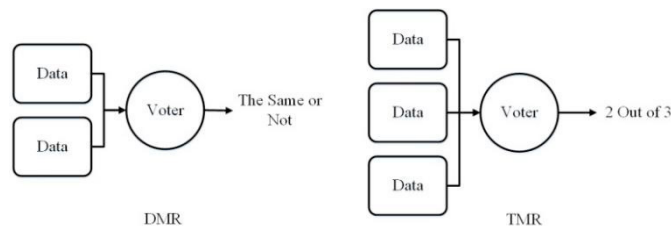


Fig. 1. DMR and TMR

2. Research Method

In this paper, Xilinx Zynq Ultrascale+ MPSoC will be used to conduct a TMR design. Because of the CCD image in the satellite becomes larger and larger, it will cost a lot of memory space to store the original size image. So, the picture is usually compressed and then it will be stored or transferred. Therefore, we carry out RGB-YCrCb conversion calculation in the processor under the background of image compression. The equation is as follows.

$$\begin{bmatrix} Cb \\ Cr \\ Y \end{bmatrix} = \begin{bmatrix} 128 \\ 128 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.5000 & -0.3313 & -0.1687 \\ -0.0813 & -0.4187 & 0.5000 \\ 0.1140 & 0.5870 & 0.2990 \end{bmatrix} \begin{bmatrix} B \\ G \\ R \end{bmatrix} \quad (1)$$

Based on the structure, there are quad-core Cortex-A53, dual-core Cortex-R5 and FPGA integrated in a single capsule. This highly integrated chip reduces chip's area massively and enriches application scenarios. In our design, quad-core Cortex-A53 and FPGA will be used. Cortex-A53 is for data calculation and FPGA is for voting. Within Cortex-A53, there are independent L1 data cache, L1 instruction cache, timer and calculation unit which fulfills four cores working independently and no data contamination. 1.2 GHz high speed calculation unit and floating-point unit can process data rapidly. In the following section, data flow, Cortex-A53's workload, FPGA voter and failure process will be thoroughly explained in sequence.

2.1. Independent Data Flow

Data flow of our TMR design is shown in Figure 2. We assume CCD image data stores in SD card temporarily and it will be compressed later. Core1-Core3 acquire data from SD card to DRAM independently which every core has their exclusive DRAM address. After the calculation, 3 AXI-4 buses will be used to transfer data to BRAM in FPGA. Every A53 core owns different BRAM address and then voters start to vote. The entire design ensures the data from three different cores work independently. In this paper, we call data from one core as one channel and there are 3 channels in the design.

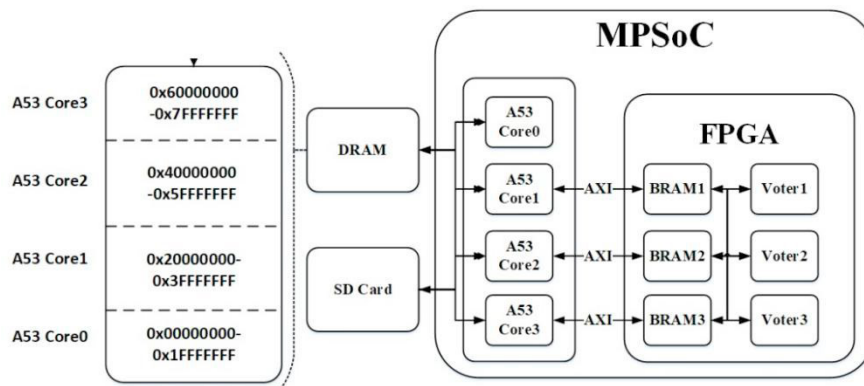


Fig. 2. TMR Data Flow

2.2. Cortex-A53's Workload

Normally, TMR needs three cores only and Cortex-A53 has four. We use the fourth core to control some steps of Core1-Core3, deal with interrupt and scrub FPGA. The other three cores work the same. The following Figure 3 shows the division of labor of the four cores.

Core0 mainly monitors the overall operation. Watchdog is implemented in the very beginning of the core to avoid the program stuck in an infinite loop. We use internal power on reset with watchdog. When time's up and the program haven't finished feeding the dog, it will reset the system. Except the error state register, all logics, registers and RAM will be reloaded. For Core1-Core3 in turn start, their program won't initiate simultaneously. In order to synchronize that three cores, Core0 will send an SGI interrupt to wakeup Core1-Core3. After the interrupt, Core0 will control the three cores to read SD card in turn or else competition will occur. The same operation will be adopted when the three cores update their address to SD card. Due to the SEU in FPGA, logics in FPGA can be wrong. Core0 also has to assure FPGA works fine by blind scrubbing at regular intervals. When scrubbing the FPGA, the Golden File will be loaded through PCAP interface to let it work again.

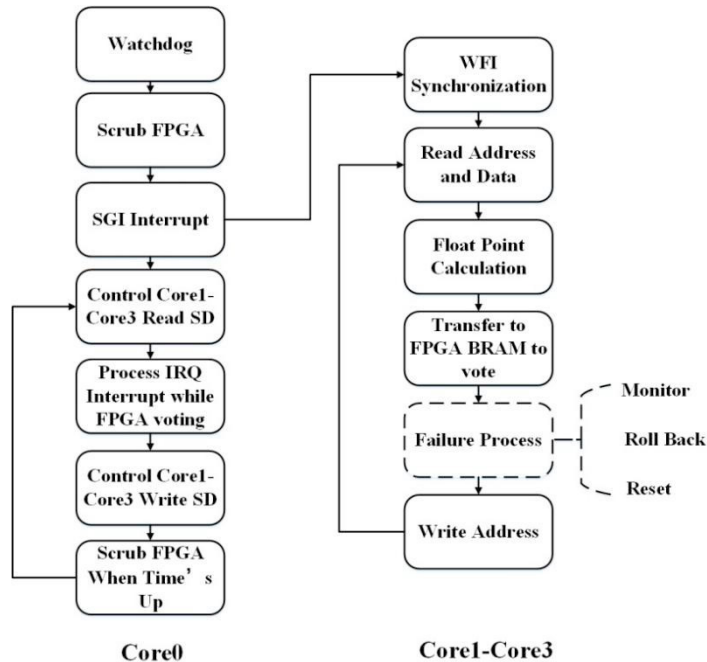


Fig. 3. Core0-Core3 Work Load Division

Core1-Core3 are consistent, so take core1 as an example. After the boot up, core1 will enter standby mode by WFI instruction actively and wait for interrupt to synchronize. Then it will read the address which is the pointer to the data file and cross exam with the other two. If they are the same, it will continue to work. If not, start from address 0. Based on the address, it will read corresponding data, conduct RGB-YCrCb conversion and send data to FPGA. When transfer is over and no FPGA interrupt, it updates the address and turns to the next loop. If there is an interrupt, it will enter failure process and has different operation based on the information provided by FPGA.

2.3. FPGA Voter

As for FPGA, the only work is to vote. If we take the fourth core as a voter, it will be only one voter in hardware and it will be easily affected by SEU. The heterogeneous chip's FPGA can achieve redundant voter easily. In our work, we use three voters to work in FPGA in parallel. Every voter will vote for the data coming from the three cores. Error occurs, three voters will send IRQ interrupt to Cortex-A53 which avoids single voter's SEU error. For FPGA has excellent parallel processing capability, three voters can work at the same time which guarantees the effectiveness. Data communication between Cortex-A53 and FPGA is through the AXI buses and the BRAM.

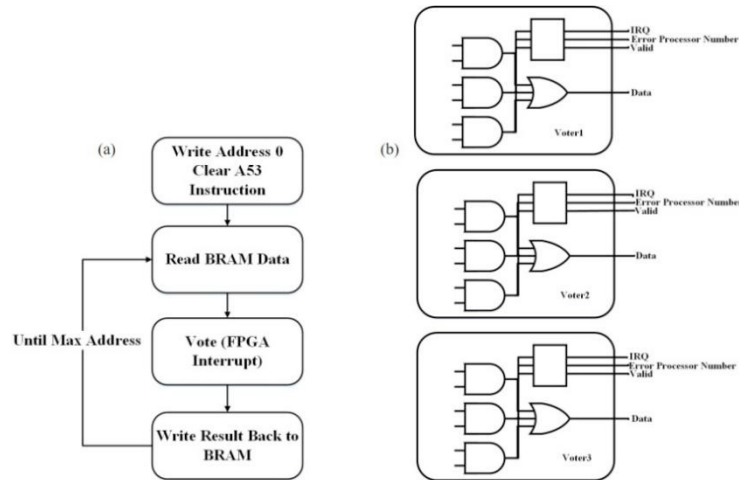


Fig. 4. (a) FPGA Flow Chart (b) Voter Diagram

2.4. Collaborative Failure Process

As for a Cortex-A53 and FPGA TMR system, data can be voted right if only one channel is wrong. However, there is still a chance if the system encounters a double channel fault or a triple channel fault due to SEU. That requires collaborative fault handling between FPGA and Cortex-A53. As long as there is an error, FPGA IRQ interrupt will be sent to Cortex-A53 to initiate failure process. The following shows how Cortex-A53 handles failure:

- If only one channel's data is wrong for limited times, TMR will correct it. Cortex-A53 will only monitor.
- If the data of the three channels are all different for only one time, TMR will vote for the data invalid and send IRQ interrupt to inform Cortex-A53 to roll back by writing the same address of this run back to the address file.
- If FPGA find it encounters massive error, the system will reboot, reload the program and update the register and DRAM. The reason why it is triggered is that when the system has a large amount of data errors in one channel, it means DRAM failure or single core failure. When the system has double channel errors more than one time, it means the data in DRAM are severely wrong due to SEU or two of the three redundant cores are malfunction due to the high energy particle.

If three kinds of fault come together, the third scenario has the highest priority and the first one has the lowest. If there is an IRQ interrupt and no error is found, the system will be reset for these two states are conflict.

3. Result and Discussion

To validate the design, we use Xilinx ZCU102 evaluation board to run our TMR model. The test is mainly verifying whether our TMR has valid SEU identification and correction ability. Therefore, we simulate the single channel error, the double channel error and the triple channel error. When inject the error, we take the single channel fault as one group and take the double channel fault and the triple channel fault as another group for the result of the last two is the same. Due to batch processing of image conversion, we calculate 64 data as one group at one transmission to mimic 8*8 image pixel processing.

From ILA waveform in FPGA side, the single channel error can all be detected and FPGA has written back the right result to BRAM. Also, FPGA IRQ interrupt is triggered and the error processor number is listed. In this case, it is core2, core2, core1 and core3 data failure in sequence. The valid signal is high when sending the interrupt, which means only one channel is wrong.

Also, the double channel error can be detected by the voter. In the ILA waveform, FPGA IRQ interrupt is sent. The valid signal is low, which means there is no right result in this vote and FPGA requires Cortex-A53 to assist. After the transaction, the address in the next run is still the same as this run which means Cortex-A53 conducts a roll back sequence and recalculates instead of monitoring. This consists with the failure process' priority setting because it has two kinds of error in this run.

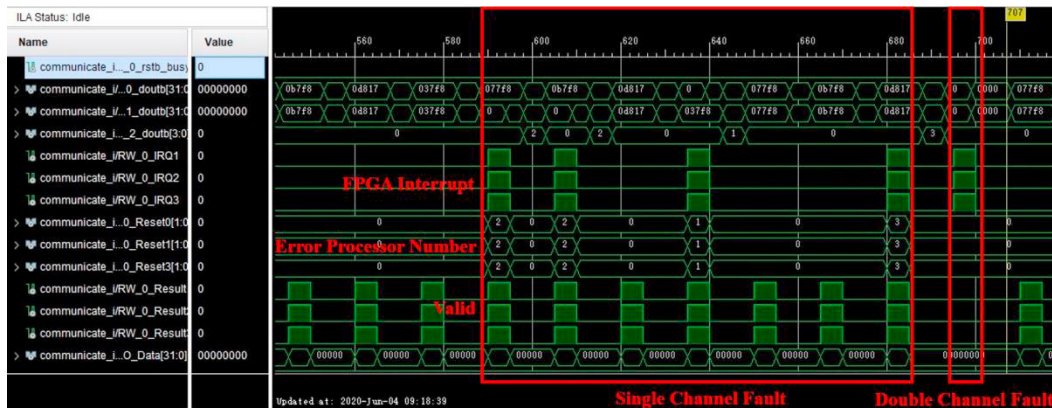


Fig. 5. ILA Result for Data Fault

The following figure shows the time consumption of SEU failure processing. Figure6(a) shows the normal run time of the program without error, which the only thing affects the time is the program we run. In this paper, we all use RGB-YCrCb conversion and it cost approximately 27.3ms. Figure6(b) shows the time, which single channel failure cost, is around 39.3ms. Figure6(c) shows the double channel failure and the processor needs 36.2ms to process. Due to the high priority, the double channel failure checks first. So, it is faster than processing single channel failure. In Figure6(d), we find that design with collaborative failure handling costs 27%-44% more time to process error which compares to the design that it only has TMR design, but without collaborative handling. The extra time is acceptable because it prevents redoing the whole project and ensure the correctness which TMR only design can solely rectify the single channel failure and needs to rerun from the beginning when encounters the double channel error.

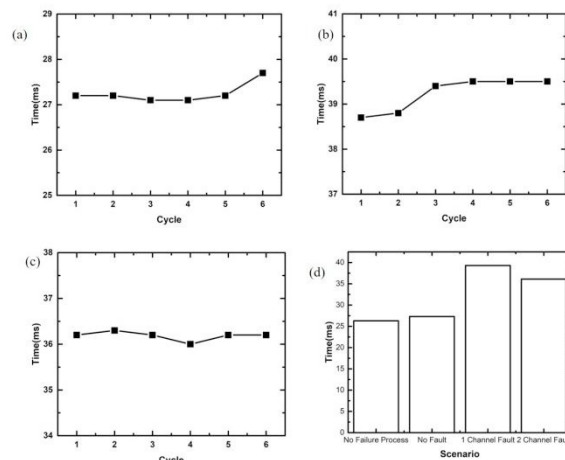


Fig. 6. (a)Time Consumption without Error. (b) Time Consumption with Single Channel Error. (c) Time Consumption with Double Channel Error. (d)Time Consumption Comparison.

From the perspective of the utilization of various resources of TMR design, LUT, FF and BRAM are all less than 6% utilization rate, and power is also relatively low. If the voter can be solidified by custom high-speed ASIC, it can be assembled as an ARM+ASIC microsystem which can further improve operation speed. In further study, we will focus on fault injection based on the different space radiation level and thoroughly study massive, efficient and precise fault injection and evaluation method on heterogeneous processors with collaborative TMR design.

Table 1. Resource Utilization

Resources	Utilization
LUT	15755 (5.8%)
FF	17120 (3.2%)
BRAM	31 (3.4%)
FPGA Power	0.818W
A53 Power	2.856W
Total Power	3.674W

4. Conclusion

Collaborative TMR system with watchdog, rolling back and scrubbing technique has excellent fault identification and correction ability. With the development of the semiconductor industry, more and more highly integrated heterogeneous processor emerges. By system level collaborative TMR reinforcement, this kind of novel processor can mitigate SEU aggressively and weaken the effect of high-energy particles. Therefore, commercial heterogeneous chip with collaborative TMR has a promising future in commercial spaceflight and it is significant to study system-level radiation harden technology for space application.

References

1. Eng. Dragoş-Ronald Rugescu, L.A.V., Gracefully Degrading Triple Modular Redundancy in FPGA Design with Application to Harsh Radiation Environments. 2016 15th RoEduNet Conference: Networking in Education and Research, 2016.
2. Peña-Fernández, M., et al., Dual-Core Lockstep enhanced with redundant multithread support and control-flow error detection. *Microelectronics Reliability*, 2019. 100-101.
3. Zhou, B., et al., Research and Implementation of Spacecraft On-Board Software Maintenance Based on TMR System, in 2018 Eighth International Conference on Instrumentation & Measurement, Computer, Communication and Control (IMCCC). 2018. p. 366-369.
4. Cui, X., et al., Fault-tolerant method for anti-SEU of embedded system based on dual-core processor. *The Journal of Engineering*, 2019. 2019(23): p. 8755-8759.
5. Tabero, J., et al., Modular fault tolerant processor architecture on a SoC for space. *Microelectronics Reliability*, 2018. 83: p. 84-90.
6. Iturbe, X., et al., A Triple Core Lock-Step (TCLS) ARM® Cortex®-R5 Processor for Safety-Critical and Ultra-Reliable Applications, in 2016 46th Annual IEEE/IFIP International Conference on Dependable Systems and Networks Workshop (DSN-W). 2016. p. 246-249.
7. Poupat, J.-L., TCLS ARM for Space. *DATA System In Aerospace*, 2016.
8. G.Kahe, Triple-Triple Redundant Reliable Onboard Computer Based on Multicore Microcontrollers International Journal of Reliability,Risk and Safety, 2018. 1.
9. Sha, W.Z.L.Y.C.W.L., Design for fault- tolerant boot loading system for on- board embedded software. *Electronic Design Engineering*, 2019. 27(8).
10. Yu, S., et al., High-Performance and Energy-Efficient Fault Tolerance Scheduling Algorithm Based on Improved TMR for Heterogeneous System, in 2018 IEEE Intl Conf on Parallel & Distributed Processing with Applications, Ubiquitous Computing & Communications, Big Data & Cloud Computing, Social Computing & Networking, Sustainable Computing & Communications (ISPA / IUCC / BDCloud / SocialCom / SustainCom). 2018. p. 207-214.
11. Yoon, J. and H. Kim, Time-redundant recovery policy of TMR failures using rollback and roll-forward methods. *IEE Proceedings - Computers and Digital Techniques*, 2000. 147(2).
12. Kucera, P., et al., Realibility model of TMR system with fault detection. *IFAC Proceedings Volumes*, 2006. 39(21): p. 468-472.
13. R.E.Lyons;W.Vanderkulk, The Use of Triple-Modular Redundancy to Improve Computer Reliability. *IBM Journal*, 1962.
14. Luis Alberto Contreras Benites, F.L.K., Automated design flow for applying Triple Modular Redundancy (TMR) in complex digital circuits. 2018 IEEE 19th Latin-American Test Symposium (LATS), 2018: p. 1-4.
15. Huichu Liu, Matthew Cotte, Suman Datta, V.N., Soft-Error Performance Evaluation on Emerging Low Power Devices. *IEEE Transactions on Device and Materials Reliability*, 2014. 14(2): p. 732-741.

16. Wood, A.D.A., The Impact of New Technology on Soft Error Rates, in 2011 International Reliability Physics Symposium. 2011.
17. P. Hamcha, T.K., J. Maiz, S. Walstra, B. Bloechel, J. Tschanz G. Dermer, S. Hareland, P. Armstrong, S. Borkar, Neutron Soft Error Rate Measurements in a 90-nm CMOS Process and Scaling Trends in SRAM from 0.25-um to 90-nm Generation, IEEE International Electron Devices Meeting. 2003.
18. Jia Liu, W.Y., RuiTao Zhang, XiaoGang Feng, Yuxin Wang, Guangbing Chen, Ruzhang Li, Yao Li, Jing Yang, Dongbing Fu. Development of a Radiation-Hardened 0.18 um CMOS Standard Cell Library for Space Applications in 2014 IEEE International Conference on Electron Devices and Solid-State Circuits. 2014. Chengdu.
19. Petrović, V., G. Schoof, and Z. Stamenković, Fault-tolerant TMR and DMR circuits with latchup protection switches. *Microelectronics Reliability*, 2014. 54(8): p. 1613-1626.
20. Nadim F. Haddad, R.D.B., Richard Ferguson, Andrew T. Kelly, Reed K. Lawrence, Daniel M. Pirkel and John C. Rodgers. Second Generation (200MHz) RAD750 Microprocessor Radiation Evaluation. in 2011 12th European Conference on Radiation and Its Effects on Components and Systems. 2011. Sevilla.